

REVIEW ARTICLE

The optical fibre Sagnac interferometer: an overview of its principles and applications

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REVIEW ARTICLE

The optical fibre Sagnac interferometer: an overview of its principles and applications

B Culshaw

Department of Electronic and Electrical Engineering, University of Strathclyde,
Royal College Building, 204 George Street, Glasgow G1 1XW, UK

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Abstract

The optical fibre Sagnac interferometer was first demonstrated over 25 years ago. Immediately its potential for gyroscopic measurements became apparent and since the first demonstration substantial research and development investment has evolved a diversity of rotation measuring instruments. The fibre Sagnac interferometer has, however, also ventured into unexpected domains. The fibre loop mirror has become the ubiquitous reflector. Sagnac-based intruder alarms, hydrophones, geophones and current measuring systems have emerged. The Sagnac interferometer has expanded from the rotation measuring instrument into a very versatile sensing tool. Indeed, it is arguably the most successful of optical fibre sensing technologies. In this paper, we review both the principles and applications of the fibre Sagnac interferometer. The background theory highlights the need to understand the conditions for reciprocity within the interferometer network. The applications range from the expected gyroscopes into novel hydrophone arrays and intruder detection systems. The basic interferometer is now well understood and the engineering required to realize useful and effective instruments has been carefully defined. Its versatility though continues to amaze even the most experienced practitioner and doubtless its applications potential will continue to expand.

Keywords: fibre optic sensors, fibre optic gyroscope, Sagnac interferometer, navigation system

1. Introduction

The Sagnac interferometer was first demonstrated by Monsieur Sagnac in 1913 [1]. The basic idea is very simple and it was a great tribute to Monsieur Sagnac's experimental ingenuity that he managed to make it work at the time. Figure 1 shows the principle. Light from an optical source is injected through a beam splitter such that it traverses in two directions around a loop. If that loop is rotated then light propagating in the same direction as the rotation has to 'chase' the beam splitter whereas that opposing the rotation finds the beam splitter advancing towards it. Consequently, the two directions within the rotating loop see slightly different optical paths. This

difference is detected using optical interferometry. Sagnac's original experiment used a loop about 1 m square and detected fringes when the loop rotated at a few times a second.

The Sagnac interferometer therefore detects rotation. The question has to be—especially in the centenary year of Einstein's annunciation of marvellous physics—'rotation relative to what?' The answer is that ill-defined quantity 'inertial space'. In other words, the Sagnac interferometer measures rotation with respect to the fixed stars in the galaxy rather than for example with respect to the (rotating) Earth's surface.

Measuring rotation is an essential part of virtually all navigation systems and many surveying systems. For example,

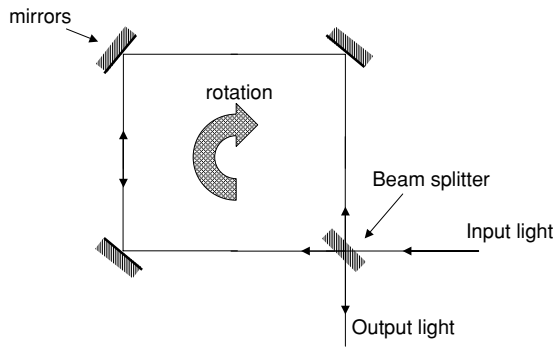


Figure 1. The Sagnac effect in air: the clockwise path sees the beam splitter receding.

if we wish to map out direction on the Earth's surface, all we need to do is take three gyroscopes with their sensitive axes precisely oriented on orthogonal axes and measure the component of the rotation detected on each of these axes. These components will add up to the 15° per hour rotation rate of the Earth which itself is oriented in a pure North–South direction. Consequently, the orientation of the coordinate system with which the gyroscope is aligned with respect to the Earth's axis can be accurately defined. To complete the measurement, we need to also know the latitude at which we are located which in turn gives the component of the Earth's rotation rate in a plane vertical to the Earth's surface. Figure 2 shows that the principle of this so-called gyro compass and gyroscope clusters are frequently used in this mode. In figure 2, Ω_E is the (total) rotation rate of the earth ($15^\circ/\text{h}$) and Ω_{ENS} , Ω_{EEW} and Ω_{EVERT} are components of Ω_E measured at latitude θ in the horizontal North–South, horizontal East–West and vertical directions, respectively. Simple trigonometry can then relate the positions of the three measurement axes with NS, EW and vertical directions at θ . Vehicle position after setting the initial orientation can also be readily derived from gyroscope measurements operating in conjunction with accelerometers. Precisely defined starting direction before the

vehicle moves (the gyro compassing stage) orients the initial trajectory. Thereafter, gyroscopes continue to map direction and the readings from accelerometers are integrated twice to find position going along three mutually orthogonal axes [2]. These inertial navigation systems kept our aircraft precisely located for many years prior to the introduction of readily available GPS and are still used as back up. Additionally submarines continue to plot their whereabouts using this technique and a host of other systems use gyroscopes to maintain vehicle stability and to calculate precise azimuth. The gyroscope is then a universal tool with many applications.

The Sagnac effect remained a dormant curiosity until the invention of the laser. In the early 1960s Rosenthal, Macek and Davis, and Aronowicz began to appreciate [3, 4] the potential of the ring laser to measure rotation. The ring laser gyroscope appeared as a product a couple of decades later and continues to be regularly utilized in aircraft and submarine navigation. The ring laser is an active system and uses the fact that the laser cavity has a slightly different effective optical length in two directions and therefore resonates at slightly different frequencies. Measuring the beat frequency and knowing the optical wavelength gives a direct reading of rotation rate [4]. The operation of the ring laser gyroscope is complicated by the fact that the two beams in the active cavity scatter into each other and therefore couple mutually locking the oscillators at low rotation rate. This is compensated by introducing a randomized 'dither' by mechanically rotating the gyroscope. The ring laser gyro was first used in commercial aircraft with the introduction of the Boeing 757.

The optical fibre version (figure 3) of the Sagnac interferometer gyroscope uses an external source and in effect simply enables the single turn loop in Sagnac's original experiment to become a multiple turn loop, thereby extending the optical delay and enhancing sensitivity. The fibre optic gyroscope (FOG) was first demonstrated by Vali and Shorthill [5] in 1976. Following the inevitable long development period for precision instruments, the FOG was first used on the Boeing 777 when it was introduced in the mid

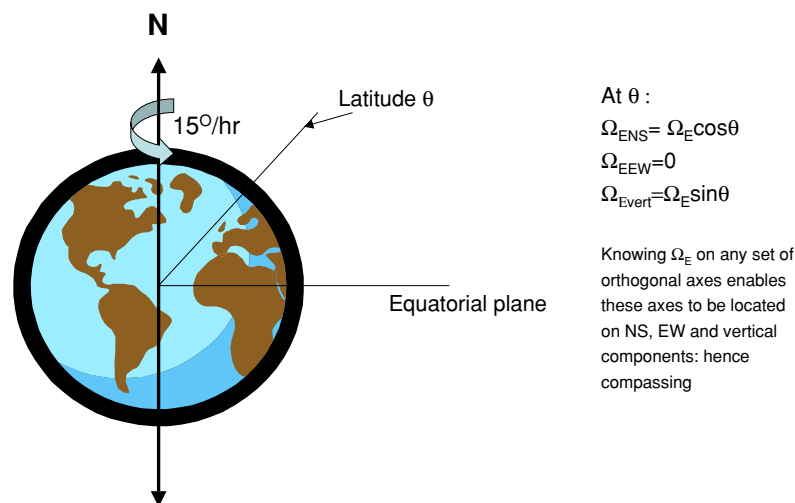


Figure 2. The principles of gyro-compassing.

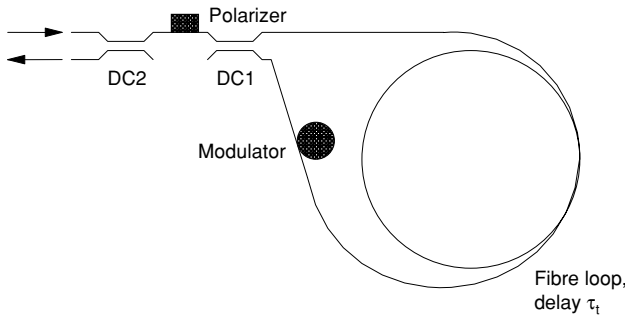


Figure 3. The fibre optic Sagnac interferometer in the minimum configuration architecture.

1990s—though unlike the ring laser which was used for inertial navigation the FOG initially found its niche in the aircraft stabilization and heading control systems. However, as we shall see, in the decade or so since the FOG's first introduction as a commercial entity, very substantial progress has been made and sensitivity and accuracy potential surpassing that achievable by competing rotational measurement technologies has been demonstrated [22]. Additionally the FOG has no moving parts and is therefore mechanically extremely rugged. Furthermore, its dependence on highly precise assembly techniques is relatively low and consequently its inherent cost is extremely competitive. The FOG is arguably the great success story for fibre optic sensor technology. Actual total annual sales of units are difficult to estimate but figures in the several hundred thousand appear reasonable. Many companies in the US, Japan and Europe rely significantly on the FOG as a mainstay of their product line.

The fibre Sagnac interferometer—like any Sagnac interferometer—detects more than simply rotation rate. Indeed, it detects any non-reciprocal phenomena which occur as the light traverses the loop. Consequently, time varying physical effects and inherently non-reciprocal phenomena such as Faraday rotation can also produce a fringe shift. This is both a blessing and a curse. As the former, it has resulted in the Sagnac interferometer finding applications in current detection (Faraday rotation) and in dynamic acoustic detectors (hydrophones and geophones in particular). The latter imposes constraints on packaging and detection coil geometries to minimize the effects of magnetic fields, acoustic fields and thermal transits.

The remainder of this review is divided into four sections which start by recounting the basics of the fibre Sagnac interferometer concentrating, in particular, on the necessary conditions to maintain reciprocity between the two loop directions and additionally to detect small deviations in reciprocity to ensure the maximum sensitivity. Then we move on to the fibre gyroscope which remains the most common manifestation of the fibre Sagnac interferometer. Thereafter, we continue to discuss current sensing and acoustic sensing. The treatment which we can give in a short paper will of necessity be brief and relatively superficial. For those interested in more details there is Hervé Lefèvre's excellent text [6], Robert Smith's compilation of papers [7] and a series of conference proceedings which have been dedicated in part or in full to the topic of fibre gyroscopes [8].

2. Fibre optic Sagnac interferometers: the basics

2.1. The Sagnac effect in optical fibres

It is straightforward to deduce from figure 1 the phase difference between the two directions of propagation in the Sagnac interferometer in which the light travels in air (or more strictly vacuum). If the time taken for the light to traverse the loop path is Δt then within that time the beam splitter travels a distance Δx :

$$\Delta x = \Omega R \Delta t \quad (1)$$

where R is the radius of the circle which forms the loop and Ω is the rotation rate in radians per second.

The actual path difference is twice this amount so with a little manipulation we find that the phase difference ϕ between the two beams is

$$\phi = \frac{8\pi A \Omega}{\lambda_0 c} \quad (2)$$

where A is the area enclosed by the loop (which can now be of any arbitrary shape without affecting the generality of the expression).

Of course, this strictly speaking applies only to the case of the transmission of light around the loop in vacuum. If we replace the vacuum path with the optical fibre as indicated in figure 3 then we slightly modify the velocities of light in each direction to

$$c_{\pm} = \frac{c}{n} \pm \Omega R \alpha \quad (3)$$

where the suffix '+' applies to propagation with the direction of rotation and α is the relativistic drag imposed by the dielectric medium given as $\alpha = 1 - (1/n^2)$ with n being the effective index of the fibre forming the loop.

Interestingly, if we substitute the delays as a function of velocities as before, we still end up with exactly the same expression for the phase delay in terms of rotation rate as that given in equation (2) [9]. Figure 4 shows this dependence for typical gyroscope dimensions normalized in terms of L , the length of the fibre and R , the radius of the loop around which the fibre is wound. The principal features of this graph are that for rotation rates of navigational significance (around 10^{-2} deg per hour) sub microradian phase shifts are to be expected. Consequently, the paths must be exactly balanced in the two directions. As we have already mentioned, effects other than Sagnac rotation induced phase can in fact produce significant path differences. This length then leads to another of the important basic considerations in fibre optic gyroscope design—Lorenz reciprocity.

2.2. Lorenz reciprocity in the fibre optic gyroscope

The basic principle of Lorenz reciprocity is extremely simple. It applies to linear, time invariant four port networks [10]. For optical fibre, the four ports correspond to the inputs along the two orthogonally polarized axes of the fibre (figure 5). The propagation phase in the two directions is exactly identical if the input and output polarization states are *exactly* identical. The definition of polarization state here includes both the direction of polarization and the spatial mode. Figure 5 illustrates that if the input and output spatial modes are not exactly identical and if there are coupling points within the

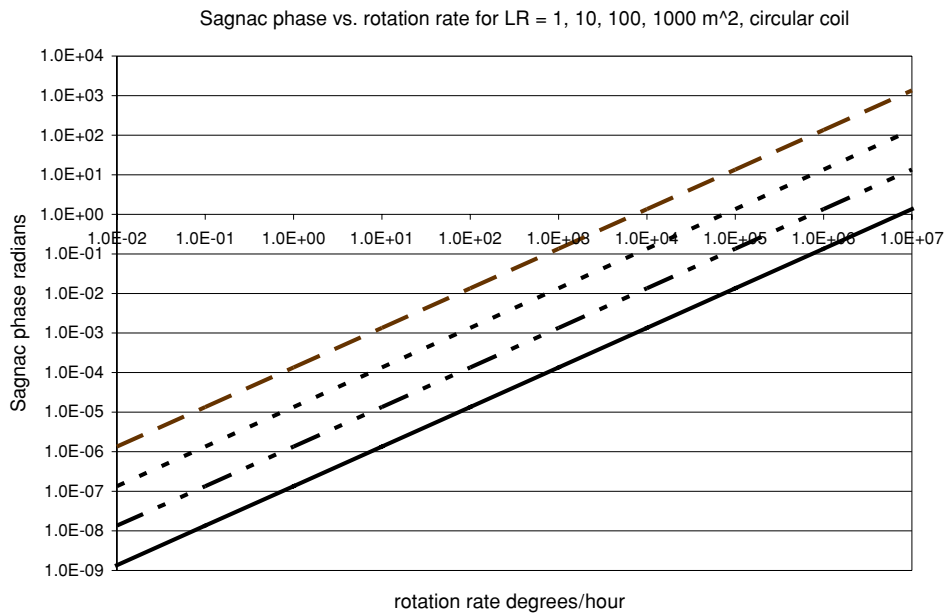


Figure 4. Values of the phase difference due to rotation in a fibre Sagnac interferometer: $\lambda = 1.5 \mu\text{m}$; LR = 1 (lowest), 10, 100, 1000 m²: LR = 100 m² is typical.

- phase same in each direction if:
 - time invariant
 - no non reciprocity e.g. Faraday effect
 - Linear propagation (*Kerr effect*)
 - same spatial mode at each end (*see across and text*)
- at 10 deg/hour Sagnac phase is $<10^{-13}$ of total @ 1.5 microns, 1km fibre length

2 spatially orthogonal modes X and Y,
one coupling point, C, along fibre length:

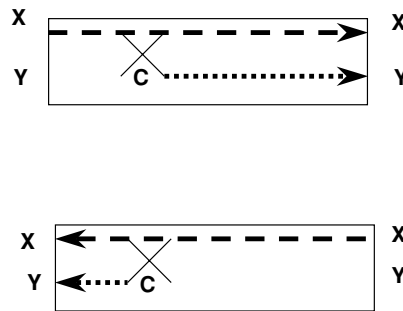


Figure 5. Lorenz reciprocity for a four port network (see the text).

fibre path between the two modes (as inevitably is the case in practice) then in general the phase paths in the two directions will not be identical.

In order to visualize how these phase discrepancies may arise consider light launched in mode X in both directions along the fibre path and consider a single coupling point between modes X and Y which is orthogonal to it. Remember too that the output phase is the weighted phasor sum of the X and Y components in each direction. Clearly there will be a *difference* in phase delay after coupling for mode Y in each of the two directions. Accordingly, the weighted phase of X and Y added together will be different in each direction dependent upon the position of the coupling point and its strength. Values for the propagation constant differential between the X and Y states can vary from approximately one part in 10^{-7} for communication grade fibre to 10^{-4} or even 10^{-3} in some forms of deliberately birefringent fibre. To put some perspective on this, at 10° per hour the Sagnac phase (figure 4) is less than one part in 10^{13} of the total length of a 1 km fibre.

One way to eliminate this mode coupling effect is to incorporate a perfect polarizer which totally removes the influence of state Y even after the coupling process. However, such perfect polarizers are fundamentally impossible. To obtain some perspective on this (figure 6), the minimum detectable phase in a well engineered Sagnac interferometer is of the order of 10^{-7} rad. Leakage of the Y component in an orthogonal phase to the X component will produce a 10^{-7} rad apparent phase distortion if the amplitude of Y is 10^{-7} of the amplitude of X. In other words, if we do assume equal amplitudes in each of the X and Y states, the rejection ratio of the polarizer should be 140 dB. This simply does not happen so alternative approaches are essential and these are discussed in more detail below.

Additionally the paths in the two directions must be exactly identical so that the output port must be taken at the second directional coupler as indicated in figure 3. Looking at the output from the free port of the first (loop coupler), we soon see that in the clockwise direction the light never traverses the coupler whereas in the anticlockwise direction the light must

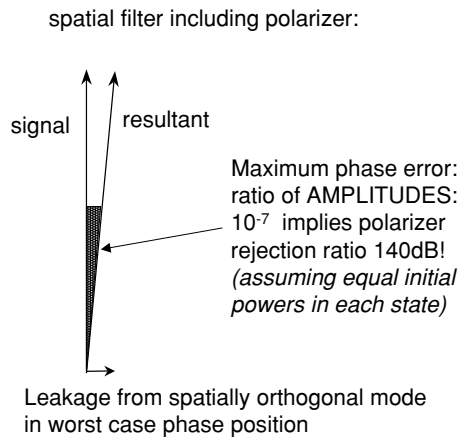


Figure 6. Phase distortion in resultant from vector sum of coupled (unwanted polarization state) and primary waves.

traverse the coupler twice, thereby breaking the fundamental basis of Lorenz reciprocity (and inevitably introducing a variable and unpredictable offset in the interferometer fringe pattern). As an indicator of these differences, recall that in a perfect, loss-free directional coupler the phase differential between cross and through ports is exactly $\pi/2$. Minute losses modify this when compared to the necessary detection tolerances.

There are other assumptions here. First and foremost the medium through which the light propagates must be linear—that is its refractive index should not vary with optical field strength. In practice [11] this is never the case. Specifically, in optical fibres the optical Kerr effect can be significant for fibre gyroscopes even at very low power densities [12].

Furthermore, the medium must be time invariant such that the condition of the medium at a particular point for light travelling in one direction is exactly identical to that for the same light travelling in the other. Should there be a change during the time delay then this will result inevitably in an apparent phase change between the two directions of propagation. In practice, the delay round the loop could occupy periods of the order of microseconds and small changes in pressure and/or temperature can easily occur within this time interval. Remember that the required sensitivity is below the microradian level and that in a path length of 100 m (which is typical, indeed perhaps rather low for a gyroscope) the phase delay is in the region of 10^9 rad. In other words, we are looking for stability of one part in 10^{15} or better. The phase delay in fibres changes by ten parts per million per degree centigrade so maintaining this degree of balance requires that temperature should not change by more than 1 nanodegree or thereabouts—a tall order. Similarly small changes in local pressure due to acoustic waves or vibration occurring within the transit time can also produce apparent offsets.

Finally, there are other genuinely non-reciprocal effects of concern, most notably the Faraday rotation introduced by magnetic field. Again very small magnetic field values can be detected using the Sagnac interferometer—or alternatively expressed stray magnetic fields can cause apparent rotational offsets.

The fibre gyroscope is then a somewhat delicate beast and requires careful engineering to ensure that conditions for Lorenz reciprocity are indeed met.

2.3. Multipath effects: linear deviations from Lorenz reciprocity

The reciprocity conditions in optical fibre Sagnac interferometers are very stringent and small imperfections can cause significant interfering effects. An indication of this can be gleaned from figure 6 which illustrates interference between the desired modes in the spatial filtering system of the Sagnac interferometer with a small amount of uncontrolled undesired mode introduced, for example by polarization coupling. We have already observed that the required sensitivity for the Sagnac interferometer is in the region of 10^{-7} rad. Consequently, the phase distortion introduced by spurious modes must be less than, or comparable to this value. In the worst case, the interfering mode is orthogonal in phase to the desired one so a relative amplitude of a mere 10^{-7} is sufficient to cause troublesome phase distortion. In other words, purity of the spatial mode filter should be such that it is capable of rejecting signals 140 dB below the carrier. Such filters simply do not exist so finding a viable approach to achieving this equivalent performance is critical in any Sagnac interferometer design. A combination of randomizing the interfering signal and utilizing appropriate signal processing algorithms (section 2.6) in the detection system does, in practice, produce the necessary effective rejection ratio when combined with suitable spatial mode filters (recalling that the spatial mode is defined through *both* the actual spatial mode pattern and its polarization). The linear contributions to this phase deviation comprise combinations of Rayleigh backscatter introduced within the fibre and polarization mode crosstalk. Backscatter introduces direct interference between forward and counter propagating beams, whereas polarization mode coupling in forward scatter tests the efficacy of the spatial mode filter. In both cases, interfering amplitudes must be kept below the magic 10^{-7} with respect to the principal interfering beams.

Rayleigh backscatter in optical fibres is well known. Indeed, it is utilized most effectively in the optical time domain reflectometer as a test probe for the performance of optical fibre links. For a fibre with a backscattering loss α dB per kilometre, the captured backscatter B is

$$B = \frac{NA^2}{4\pi} \left[\frac{\alpha \cdot dl}{2.302} \right] \quad (4)$$

where NA is the numerical aperture of the fibre and dl is the length over which the backscatter is collected. Inserting typical figures into this relationship, taking 1 dB per kilometre and $NA = 0.1$ we find that for the total backscatter fraction to be less than 10^{-14} (remember we are looking at power), dl should be less than a number of the order of 10^{-11} m. Clearly this is not going to happen. In effect we are saying that the backscatter from a few molecules of silica is sufficient to cause an interfering signal so other approaches are essential. There are in practice two aspects to this. The first is to minimize the source coherence length which will in effect randomize most of the backscatter. The second is to utilize suitable signal processing schemes. The basic approach in these signal processing schemes is to utilize the observation that the delay line in the fibre loop is effectively a dual tap finite impulse response filter which operates on the modulation imposed at one end of the loop (figure 3). This modulation is

delayed by τ_i on the clockwise light with respect to counter clockwise light. If the influence of the back scatter signals when detected at the modulation frequency is configured by the delay line to add to zero, then there will be no net impact from the backscatter phenomena. The minimum coherence length that we can obtain is of the order of a few microns so that the signal processing needs to compensate for about two additional orders of magnitude in amplitude and it turns out (see section 2.6) that this is indeed feasible.

The polarization story is similar. Most optical fibre Sagnac interferometers utilize polarization-maintaining fibre and again there is some scatter between two modes of the fibre determined by the so-called h parameter. This h parameter determines directly the power in the orthogonal mode which is scattered from the principal mode per unit length. In a good polarization-maintaining fibre, h is typically 50 dB m. Obtaining the magic 140 dB therefore requires that the maximum polarization scattering length is 10^{-9} m—clearly not an option. However, we should recall that once again all that matters is the coherent scatter which for a short coherence length source operates over a few micrometres. 50 dB m is equivalent to 110 dB μm so for a coherence of say 10 μm . The coherent light coupled into the unwanted polarization mode is at -100 dB. Consequently, we need to only gain 40 dB through other filtering mechanisms and all is well. 40 dB polarizers and much better are feasible so the polarization scatter between modes is effectively minimized. There are additional benefits too in highly birefringent fibre since forward scatter also becomes incoherent after a few beat lengths of optical fibre.

This discussion has demonstrated that the linear cross coupling effects between propagating modes, whether due to forward scatter and polarization cross coupling or Rayleigh backscatter, can indeed produce significant interference to the principal mode which is used in the Sagnac interference process. The amplitude of these linear interfering phenomena cannot be kept below that necessary to achieve the required modal selectivity. Consequently, other approaches are essential and the use of the low coherence source is central to the operation of a fibre Sagnac interferometer. The interferometer functions on the zero path difference fringe so the coherence length of the source need be no more than single numbers of wavelengths. This is an extremely powerful benefit of the fibre Sagnac interferometer and automatically can eliminate many of the spurious reflection effects which plague other interferometry systems which rely upon high degrees of temporal coherence in the optical source.

2.4. Nonlinearities: the Kerr effect

The optical Kerr effect is present in all materials including relatively stable and passive substances such as the silica which is the major component of optical fibres. In silica, the Kerr coefficients are very small [11] so it is tempting to argue that the influences of nonlinearities could be trivial at the low power levels—typically in the microwatts region—which are used in most optical fibre Sagnac interferometers.

The Kerr coefficients are the second-order term in the expansion of the polarization experienced in a material as a function of electric field. In other words, the refractive

index has a small correction which is proportional to the intensity of the electromagnetic wave. The situation in the fibre Sagnac interferometer is that not only do we have the effect of self-induced correction to the refractive index, we also have the influence of the light wave coming in the opposite direction. If we look at the case of a continuous wave of constant power I_1 meeting another wave of constant power I_2 propagating in the opposite direction, we find that the change in propagation constant $\Delta\beta$ of I_1 is

$$\Delta\beta_1 = C [I_1 + 2I_2] \quad (5)$$

where C is a constant. Similarly for I_2

$$\Delta\beta_2 = C [I_2 + 2I_1]. \quad (6)$$

These equations tell us that the self-induced Kerr effect is half the cross effect [12]. If the two intensity levels I_1 and I_2 are exactly equal then the change in propagation constants in the two directions is exactly the same and the effect is compensated. In practice, this is difficult (impossible?) to ensure since small changes in the coupling coefficient of loop directional coupler with temperature, for example, will shift the balance and produce an apparent drift. To reiterate, we are looking at sensitivities to path differences of the order of parts in 10^{-17} at the detection limit. Alternatively, we could conceive a modulation scheme in which the self-effect is constant, whilst the cross effect is effectively halved by, for example, having an exactly 50% duty cycle pulse modulation on the source. However, this is again extremely difficult to do with sufficient accuracy so is impractical.

There is though a viable way around this. We express the difference in propagation constants introduced through the Kerr effect—that is in effect a spurious rotation rate Ω_k dependent on the Kerr effect—by looking at the difference between the two Kerr effect propagation constants in equations (5) and (6) and putting these into equation (2). We obtain [12], where K is the coupling coefficient of the input directional coupler and I_0 is the time varying (with t) power input to the loop coupler:

$$\Omega_k = B(1 - K) \left[\frac{\langle I_0^2(t) \rangle - 2\langle I_0(t) \rangle^2}{\langle I_0(t) \rangle} \right] \quad (7)$$

where B is a constant. Clearly for Ω_k to be identically zero we need to ensure that the statistics of the source ensure that the term in the large square brackets itself becomes zero. A 50% duty cycle bears exactly this relationship. There is though another solution. Examination of the source statistics for Gaussian noise can show with a little algebra that a perfect Gaussian noise source also has the same statistical properties [13]. In practice, this points towards the use of super radiant diodes rather than lasers. Alternatively, a highly multimode laser diode can approach the same statistical limit for more than ten or a dozen individual, totally independent, oscillating lines. Clearly a mode locked laser far from approaches this limit.

We do then have here yet another reason for using a low coherence source in optical fibre Sagnac interferometers, namely that the Kerr effect nonlinearities are automatically compensated.

There are of course other nonlinearities which are present in optical fibres, most notably Raman and Brillouin scatter

[11]. For most fibre Sagnac interferometers these linearities are genuinely negligible. The intensities within the core of the optical fibre are orders of magnitude below the necessary thresholds. There have been a few trials of utilizing Brillouin scatter in particular as a sensing mechanism in the optical fibre gyroscope but to my knowledge none of these has gone beyond the laboratory curiosity [14].

2.5. Noise phenomena in the fibre Sagnac interferometer

Optimizing the signal-to-noise ratio of the final detected signal is one of the principal design objectives in the realization of any electronic or optoelectronic system. This process can be complex in practice but the principle is straightforward. In optoelectronic systems there are three principal sources of noise, namely source noise, shot noise resulting from the particulate nature of the detected signal and detector system/electronic noise. Any theoretical consideration of these noise sources almost invariably assumes ideal behaviour. In practice, it is often necessary to incorporate a 'noise figure' into the estimates but with careful electronic design in both the source power supply system and the detector circuitry, these excess values can typically be controlled to within a few dB. Accordingly, the idealized estimates can give us a very useful indication of potential system performance.

Shot noise describes the standard deviation of the rate of arrival of a flow of uncorrelated particles like photons. If the flow rate is N particles per second, then the variance σ_N^2 is given by

$$\sigma_N^2 = 2NB \quad (8)$$

where B is the detection frequency bandwidth. The rate of arrival of photons N is related to the optical power through

$$N = \frac{P_{\text{opt}}}{h\nu}. \quad (9)$$

N at $1 \mu\text{W}$ for a wavelength of $1.5 \mu\text{m}$ is approximately 7.5×10^{12} photons per second.

Source excess noise is predominantly associated with the so-called residual intensity noise (RIN). This arises because of what may be termed a random beating amongst all the frequency components within a broadband spectrum. It has been analysed in detail elsewhere [15] and applied to the fibre optic gyroscope. In effect, this produces an additional statistical fluctuation in the photon rate given by

$$\sigma_{\text{NRIN}}^2 = \frac{N^2}{\Delta f} B \quad (10)$$

where Δf is the bandwidth in the optical domain of the source. There is obviously then a condition at which the RIN noise exceeds the shot noise—the ratio of RIN to shot noise is

$$\frac{\sigma_{\text{NRIN}}^2}{\sigma_N^2} = \frac{N}{2\Delta f}. \quad (11)$$

As an example, at $1 \mu\text{W}$ received power, the RIN dominates for $\Delta f > 4 \times 10^{12}$ Hz, corresponding to a wavelength spectral width in the region of 13 nm. This simple example demonstrates that RIN often dominates performance in fibre gyroscopes.

In addition there is of course electronic noise. This is predominantly due to Johnson (thermal) noise in the load resistor (or its equivalent in a transimpedance amplifier

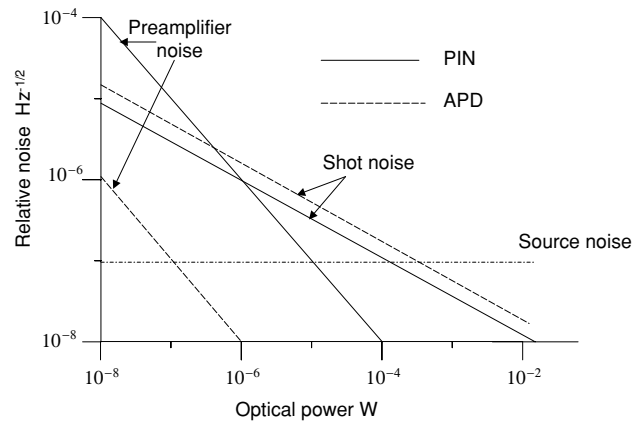


Figure 7. Relative noise sources as a function of received optical power for silicon detectors at $\lambda = 850 \text{ nm}$ (After [6] p 143).

configuration) on the photodiode detector circuit. The noise current generated is given by

$$\langle i_{\text{th}}^2 \rangle = \frac{4kTB}{R_l} \quad (12)$$

where k is Boltzmann's constant, T is the absolute temperature, q is the electronic charge and R_l is the load resistance. In practice, this electronic noise is also typically multiplied by a noise figure of a few dB.

These three relationships define the limiting performance of the fibre optic Sagnac interferometer—or indeed any other optical system. The parameter which determines noise performance is the electron (rather than photon) flow in the detector circuit so equations (8) and (11) require modification to allow for the quantum efficiency of the photo detector.

The detailed behaviour of a particular system depends very much on its individual characteristics. A typical trade off is shown in figure 7 which is based on data in [6]. Typical received powers in fibre gyroscopes are in the microwatt region and in this area shot noise limited detection is generally a good starting guideline though depending upon the source characteristics RIN can also dominate.

In terms of signal-to-noise ratio, we have already seen that $1 \mu\text{W}$ at $1.5 \mu\text{m}$ wavelength is approximately 7.5×10^{12} particles per second. Consequently on a 1 Hz bandwidth, the ratio of the mean square photon rate fluctuation to the mean photon rate is approximately 2.5×10^{-7} . Since this translates directly (through the detector quantum efficiency) into electron flow this in turn represents a good value for the achievable detector current signal-to-noise ratio. In turn, for bias on the quadrature fringe—see the next section—this gives a shot noise limited phase sensitivity of a few tenths of a micro radian in a 1 Hz bandwidth at $1 \mu\text{W}$ received power. This is a good guide to the sensitivity which is achievable in practice.

2.6. Phase detection in the fibre optic Sagnac interferometer

There are basically two approaches—open loop and closed loop. The latter uses the former as the source of the null signal to provide the feedback. Accordingly we first of all look at open loop detection.

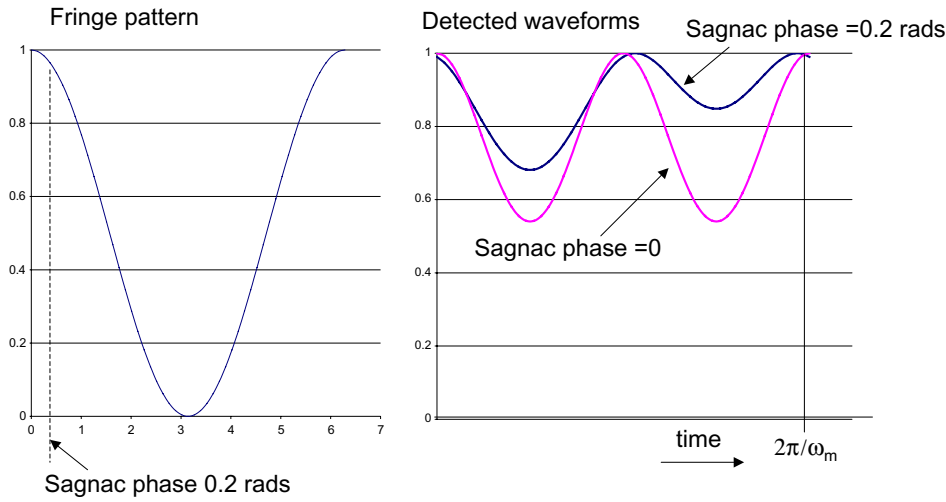


Figure 8. Open loop detection using an off centre phase modulator (as indicated in figure 3 and see equation (14)).

The basic principle is shown in figure 8. The phase modulator shown in figure 3 clearly modulates one direction of rotation immediately after its departure from the input directional coupler and the other direction immediately before it re-enters the input coupler. Accordingly there is a time difference in the phase modulated signal of approximately τ_t , the loop transit time. The phase modulator can provide a whole host of functions but the simplest is sinusoidal:

$$\phi(t) = \phi \cos \omega_m t. \quad (13a)$$

The differential phase modulation is simply $\phi(t) - \phi(t - \tau_t)$ which is

$$\begin{aligned} \phi_{\text{diff}} &= \phi_0 (\cos \omega_m t - \cos \omega_m (t - \tau_t)) \\ &= \phi_0 \sin \left(\frac{\omega \tau_t}{2} \right) \cdot \cos \omega \left(t - \frac{t - \tau_t}{2} \right). \end{aligned} \quad (13b)$$

The differential phase modulation is accordingly a maximum at the so-called loop transit frequency which is $f_t = 1/2\tau_t$. Operating at the so-called loop transit frequency with the modulation function has huge benefits, not only in maximizing the differential phase modulation between the two directions, but also in minimizing a number of other interfering phenomena.

Figure 8 shows how the differential phase modulation manifests itself as intensity modulation on the detector. If the static phase differential between the two directions of rotation around the Sagnac loop is zero then symmetry considerations tell us immediately that the detected output comprises entirely even harmonics of the modulation frequency. However, if the static phase deviation between the two directions deviates from zero by even a small amount we very rapidly see the build up of even harmonic components. It is relatively straightforward to demonstrate [6] that if the Sagnac phase is ϕ_s the detected intensity is

$$\begin{aligned} I(t) &= A_0 \cos \phi_s (J_0(\phi_{\text{diff}}) + 2J_2(\phi_{\text{diff}}) \cos 2\omega_m t + \dots) \\ &\quad + A_0 \sin \phi_s (2J_1(\phi_{\text{diff}}) \sin \omega_m t + \dots). \end{aligned} \quad (14)$$

Typically detection systems whether for nulling or otherwise seek the first harmonic component which maximizes when $\phi_{\text{diff}} = 1.8$ rad. These relationships along with the relationships in the previous section concerning noise performance can be

brought together and used to determine signal-to-noise ratio and eventual sensitivity for the Sagnac interferometer. Typical figures give eventual phase sensitivities for ϕ_s as a fraction of a micro radian in a 1 Hz bandwidth for received power in microwatts assuming that the modulation frequency ω_m is well away from $1/f$ noise in the detection circuit. In practice, for 200 m fibre loop ω_m is approximately $10^6 \pi$ rads s^{-1} .

A quick inspection of equation (14) shows that its behaviour is highly nonlinear. Additionally, it shows that the output depends upon optical amplitude so some intensity correction is desirable. In practice, this is realized through taking the ratios of the harmonic components. In principle, sufficient information can be gleaned from J_1 and J_0 though in practice the first four harmonics are often utilized. With suitable processing this can also compensate for the slight variations in ϕ_{diff} which also affect the ratio of the harmonic components in equation (14).

Closed loop systems somehow have to set the effective ϕ_s to zero.

One possible approach is the obvious one of counter rotating the gyroscope itself though this totally defeats one of the principal benefits of the fibre gyroscope—namely its strap down operation [2]. Alternatives are therefore needed and in effect the approach is to introduce a very small frequency shift between the two directions of propagation [16]. Indeed, very simple arguments lead to the conclusion that the frequency shift must be sufficient to introduce phase differential ϕ_s during the time τ_t and this corresponds to a frequency shift, Δf_{null} , of

$$\Delta f_{\text{null}} = \phi_s / 2\pi \tau_t \quad (15)$$

when ϕ_s is expressed in radians. Δf_{null} is multivalued since any value of $(\phi_s \pm 2n\pi) / 2\pi \tau_t$ will also satisfy the same relationship where n is any integer. A dynamic range in ϕ_s extending to 2π is however in practice usually more than adequate.

How should this frequency difference be introduced? With a little thought we can see how the phase modulator could introduce this by adding an additional term to equation (13b) to introduce the phase ramp at exactly the correct rate. Frequency shifting using phase ramps—serrodyne modulation—is a well rehearsed technique [17] and with a suitably configured phase

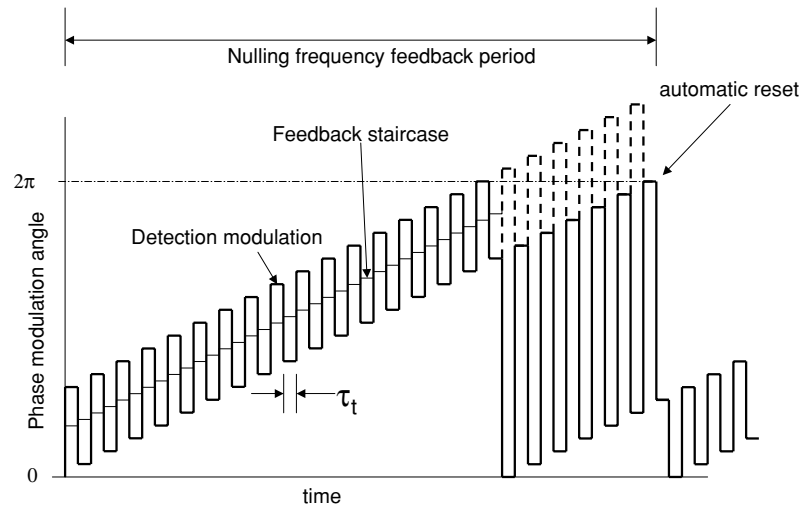
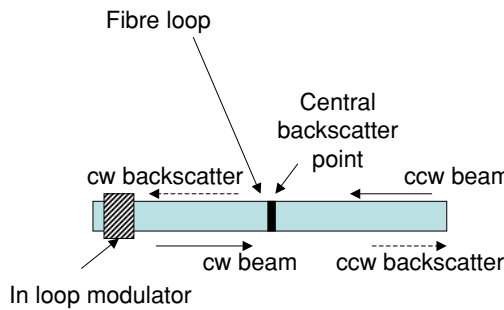


Figure 9. Staircase phase difference ramp for closed loop detection showing detection modulation superimposed (see the text).



Backscatter from cw phase modulated twice, but π out of phase (total = 0). Backscatter from ccw never modulated, so detected backscatter minimized by $f_m = 1/2\tau_t$ since detection at f_m .

Transfer function for amplitude modulation applied by the in-loop modulator:

- cosine response for source amplitude modulation;
- sine response for phase modulation

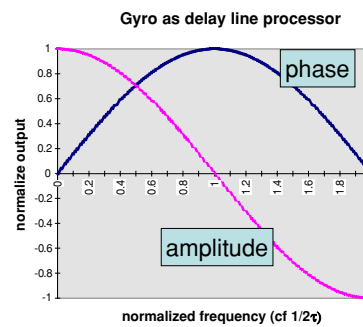


Figure 10. Backscatter and spurious amplitude modulation: filtering by use of the transit time frequency $\omega_m = \pi/\tau_t$ or $f_m = 1/2\tau_t$.

modulator introducing these phase ramps can be relatively straightforward. An important variation on this basic theme introduces a staircase [6] modulation, stepping through ϕ_s at τ_t intervals. This ‘digital phase ramp’ is an extremely powerful detection technique since it also redefines the definition of the slope of the phase ramp into counting the number of phase increments required to step the total phase differential through 2π (figure 9). There are two components in the waveform in the figure. The high frequency one is the detection modulation (at $1/2\tau_t$) applied here as a square wave. This modulation produces a square wave detected output (see figure 8). This output contains no odd harmonics of the modulation frequency ($1/2\tau_t$) if and only if the phase difference between the two arms is zero. The feedback circuit then locks this phase difference to zero by applying the necessary phase step and the steps are counted. Using a 2π phase shift detector and filtering the low frequency component of the staircase waveform resets the ramping term to zero phase. Counting the number of modulation frequency periods at reset then gives the Sagnac phase. This approach requires a very broadband modulator

system which inevitably must be constructed using integrated optic techniques [17(b)]. Additionally the feedback circuit must be carefully designed to accurately detect the completion of a full cycle of the difference frequency. The system though is enormously powerful since it operates entirely in the frequency domain on a closed loop zero detection system. The zero is in effect nulled during each detection modulation cycle. There is clearly the possibility during rapid rotational acceleration for the slope of the ramp to change during the modulation cycle. The output is then effectively an average of the rotation rate during the cycle, but with more sophisticated detection processors the rate of change can also be detected. This basic technique has found extensive application.

We have already mentioned that modulating at the transit time frequency can have benefits other than simply maximizing the differential modulation depth. Two of the most important of these are shown indicatively in figure 10. (The immense benefits of operating at the transit time frequency became progressively more apparent as the appreciation of the filtering characteristics of the loop—acting as a finite impulse response

dual tap filter—became increasingly appreciated. Much of this knowledge remained unpublished until it became apparent that the benefits were well appreciated by those in the community. They are well enunciated in far more detail than we have space for here in Hervé Lefèvre's excellent book). The first is that many phase modulators also introduce small spurious intensity modulated components at the modulation frequency. These components could obviously interfere with the detected signal. However, if we operate at the transit time frequency it becomes immediately apparent that this spurious amplitude modulation will automatically cancel provided that the transit time and the modulation frequency exactly coincide. Since at the threshold of phase detection we are looking for relative amplitudes of 10^{-6} or 10^{-7} of the fundamental compared to J_0 , then very small unwanted amplitude modulation at the modulator can be significant. The second important benefit concerns backscatter. If we consider the situation shown in figure 10, we see that the backscattered light from the centre of the fibre may also be usefully minimized, and for short coherence lengths almost cancelled by working at the correct frequency.

Figure 10 encapsulates the signal processing properties of the delay in the fibre loop between the modulation introduced by the modulator on clockwise and anti-clockwise beams. If we imagine this as an impulse response, an impulse from the modulator in the clockwise direction will arrive at the detector at say time $t = 0$, whilst that in the anti-clockwise direction would be delayed by the transit time τ_t . How this impacts upon any modulation which is applied on beams in clockwise and anti-clockwise directions depends upon whether this modulation is in phase or amplitude. The important aspects of this are that the backscatter from the centre of the loop (which is optically coherent with the forward scatter from the opposite direction) is either modulated twice in anti-phase by the modulator or not modulated at all depending upon whether the scatter originates from the clockwise beam (modulated twice) or the anti-clockwise beam. The double modulated component from the clockwise beam is, at the transit time frequency, modulated twice in anti-phase with net zero. Additionally, if the modulator introduces spurious amplitude modulation, which is almost inevitably the case for all optical phase modulators, then that amplitude modulation will be applied 180° out of phase on the clockwise and anti-clockwise beams (see cosine response in the diagram) and consequently will add to zero. Any inherent amplitude modulation will interfere with the signal which is required. Using the transit time filtering system fundamentally eliminates spurious amplitude modulation. In contrast, the phase modulation terms (see equation (13)) add constructively to produce a maximum of the desired signal when the modulation frequency is tied to the transit time. These effects are considered in much greater detail in [6–8]. The conclusion is that the benefits from operating at the transit time frequency are substantial and if at all possible Sagnac interferometers should be designed with this in mind.

2.7. The basic principles: a summary

The key to successful implementation of the fibre optic Sagnac interferometer lies in Lorenz reciprocity, the conditions for which—section 2.2—are simply stated but difficult to realize. The most basic manifestation of this is the use

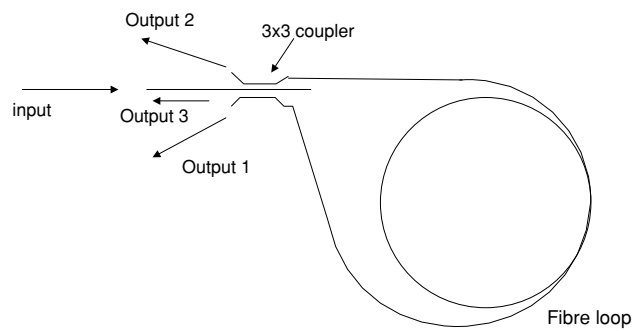


Figure 11. The 3×3 coupler architecture.

of the dual coupler and polarizer architecture shown in figure 3 which ensures that modal control is at least in principle practical. Further design features target reducing the impact of backscatter by using short coherence length sources and highly birefringent fibre and minimizing the impact of polarization conversion, predominantly using the same strategy. Fortuitously nonlinear effects are also taken care of by using a short coherence source and collectively these design features facilitate an extremely good approximation to the conditions required for total reciprocity. Thereafter, we can improve the situation yet further through careful signal processing (in particular operation at the transit time frequency) and through attention to the detail in the electronic noise mitigation and phase detection architectures. Once again, the transit time frequency plays an important part, together with synchronous detection of the interferometer output at this frequency and its harmonics.

3. The fibre optic gyroscope

3.1. Architectures

There are a wide variety of optical and electronic systems which can meet—or approximately so—the fundamental needs that have been enunciated theoretically in the previous sections. We shall briefly examine the principal optical and electronic processing options.

3.1.1. Optical architectures. The vast majority of fibre gyroscopes follow the optical minimum configuration architecture shown in figure 3. A variation on the basic theme based upon a single 3×3 coupler rather than 2×2 couplers is shown in figure 11. The 3×3 coupler is designed to split the input equally among the three output ports. The outputs from the three ports as a function of the Sagnac phase in this architecture follows a ‘three phase’ relationship shown in figure 12. Interestingly, at zero phase the outputs from the accessible ports have in the ideal case equal and opposite slopes and subtracting these outputs can be used to determine the Sagnac phase without recourse to any modulation in the loop. This arrangement is however inherently non-reciprocal since the input and output paths are different. Additionally since the coupler exhibits a finite loss, there will be a slight phase offset from the ideal positions shown in the figure. Whilst this optical architecture has been demonstrated it has not, to my knowledge, been used as a gyroscope in commercial

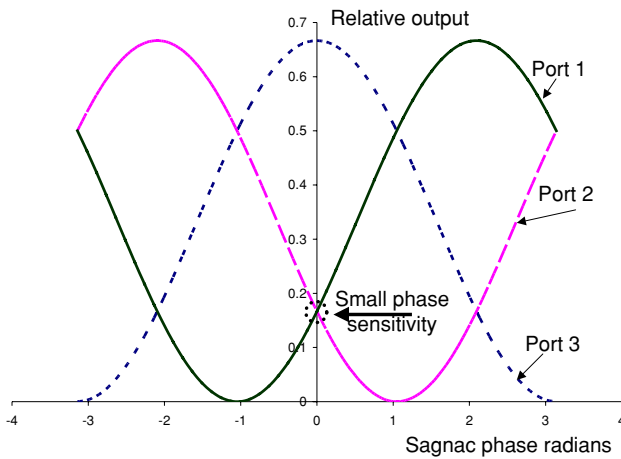


Figure 12. The outputs from ports 1, 2, 3 in figure 11 as a function of Sagnac phase.

systems. The principal practical issues include the fabrication difficulties associated with the 3×3 geometry and the inherent and unstable non-reciprocity within the 3×3 configuration. Consequently, detecting dynamic variations (see later) can be practical with the 3×3 approach.

Thus far, we have concentrated on the traditional approach to achieving Lorenz reciprocity, namely pure single mode outputs from the loop and a high rejection polarizer situated between the two couplers at the front end of the gyroscope. Additionally this architecture uses highly birefringent polarization-maintaining fibre to both retain the input polarization state and ensure that any power launched into the orthogonal state is incoherent with that launched into the primary state when it is finally detected. Using these combined approaches can give the 10^{-7} amplitude interference which is required by high performance Sagnac interferometers. There is, however, another approach which most importantly obviates the need for the polarization-maintaining fibre loop and therefore can introduce useful cost savings. This is to use depolarized light as the source [18]. The depolarized light is created by placing a Lyot depolarizer (either in bulk optic or fibre optic formats) immediately after the broadband source. This depolarized light is then injected into the polarizer. The polarizer is imperfect and so some orthogonal component is also present. However since the input light is, in principle, fully depolarized, the orthogonal components, when integrated over the detection time of the system, should add to zero, with consequently no phase distortion at the output. There is though a 3 dB loss at the input polarizer. Yet again, this input light is depolarized in the loop and in principle arrives fully depolarized back at the inter-coupler polarizer. Here the process is repeated and again the influence of the output orthogonal unwanted polarization is randomized. Both the launch into the loop and the detection at the second coupler from the loop introduce an additional 3 dB loss. Furthermore the depolarizer is never perfect and inevitably some residual polarization remains. The $10^{-7}/140$ dB effective rejection ratio persists as the performance criterion. Consequently depolarizer architectures are inherently less sensitive than the more traditional approach and the cost saving arguments whilst superficially convincing are less so when the influence of the two de-polarizers is incorporated.

The quality of the polarization-maintaining fibre is, however, the key to gyroscope performance. This fibre must maintain the state of polarization (the h value) and must also ensure that any light that does enter the orthogonal state is incoherent at the detector (the inherent birefringence or ‘beat length’ of the fibre). h values of 50 dB m are typical with beat lengths of a few (less than five) mm. Typical source coherence lengths are a few tens of wavelengths so that light scattered into the orthogonal state more than—say—a metre from the detector is totally incoherent. However, some simple arithmetic soon demonstrates that the polarizer is also necessary to remove even this small residual amount. There are many techniques for fabricating polarizers—both integrated optic and all fibre—to optimize performance [6–8].

The final major aspect of optical architectures lies in winding the coil. We have already seen that asymmetric disturbances can cause offset. In principle, this offset could be completely nulled by ensuring that the asymmetric disturbance is applied symmetrically—that is that the fibre coil is wound exactly around the mid point so that a disturbance at any point within the coil is balanced [19]. This is clearly not feasible in practice but can be approached by winding the coil with the starting point in the centre as indicated in figure 13(B) (part (a)). To avoid the fibre loops crossing at every turn, the coil has to be wound layer by layer either in the bipolar or quadrupolar format indicated. The quadrupolar format gives the best compensation though only when carefully implemented and in particular designed to have a number of layers which is exactly divisible by four (figure 13(B) (part (b))). An alternative is to randomize the winding profile. Though whilst this is somewhat less efficient at total compensation, it does significantly simplify the winding process and also has a more homogeneous response to interferences coming in from different directions [20].

In summary then, the most commonly used optical architecture is the traditional polarization-maintaining fibre and polarized source approach though depolarized architectures are occasionally utilized. Both architectures can be implemented in integrated optical or all fibre format (see the next section concerning signal processing) though the depolarized architecture is usually all-fibre. A wide variety of component designs have been incorporated into gyroscope systems. Coil winding is very important since it could and does have a major effect on the sensitivity to interference from vibration and thermal gradients. The quadrupolar winding is the ‘conventional wisdom’ of coil winding techniques for gyroscopes but the random wound coil has also found its place.

3.1.2. Open and closed loop electronics. The optoelectronic detection system has two functions. The first is to provide the modulation—preferably at the transit time frequency—from which the rotation rate can be measured. This modulation is either nulled in closed loop gyros or accurately measured in open loop gyros. The null system has the obvious benefits of immunity to drift in both electronic and optical components. The phase staircase described earlier with an added null detection square wave modulation synchronized with it is probably the most powerful technique but it does require broadband integrated optic modulators.

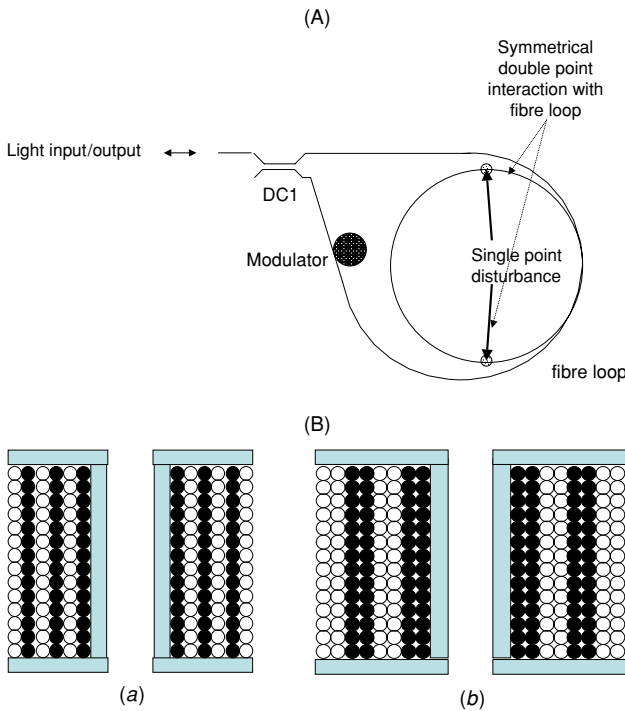


Figure 13. (A) The concept of centrally symmetrical winding to compensate for transient (thermal, acoustic) influences—the basis is that the disturbance is caused to interact with the transient simultaneously on both clockwise and anti-clockwise beams in the loop. Winding options indicated in (B). (B) Symmetrical winding in cross section—(a) bipolar and (b) quadrupolar: the coil is wound from the centre of the fibre—one side of the centre is marked (●) and the other (mirror image) (○). Note in (b) the layers are in pairs therefore the total number of layers should be a multiple of 4. In (a) the total number of layers should be in multiples of 2.

Simple piezo ceramic modulators have been very effective for open loop systems. There are, however, inevitable drift effects which need to be compensated. These originate in both the changes in effective modulation depth applied by the modulator with for example temperature and in the optical output/detection sensitivities within the optoelectronics. Again though there are compensations/correction mechanisms available almost all of which are derived from measuring the relative amplitude of harmonics of the modulation frequency including the possibility of nulling the J_2 component to define the exact value of J_1 .

There is an immense variety of optoelectronic configurations in use based upon these two principles. High precision is only obtainable using closed loop systems but the demand for lower performance gyroscopes and the cost benefits associated with removing the need for an integrated optic head end makes the open loop system attractive. Individual manufacturers have evolved their own solutions to specific application requirements.

3.2. The fibre gyroscope: some examples

There are numerous manufacturers internationally of fibre optic gyroscopes including Mitsubishi and the Japan Aviation Electronic Company, Honeywell and Northrop Grumman (formerly Litton), Ixsea and KVH. There are numerous others. The performance and details of individual products can be

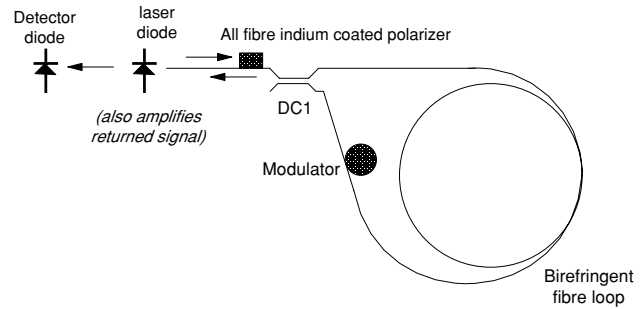


Figure 14. The 'reduced minimum configuration' gyroscope as adopted by KVH [21].

obtained by contacting the manufacturers or through their websites. The technologies involved invariably follow the basic approaches outlined above—namely open or closed loop electronics and some variation on the minimum configuration architecture. The details and the performances which can be achieved vary enormously depending upon the application and the market sector which a particular manufacturer chooses to address.

Here we shall briefly discuss two examples. KVH based in Chicago, Illinois, manufactures medium performance gyroscopes targeted towards vehicle stabilization, antenna stabilization, gyro compassing and similar areas. These are all open loop systems based upon either the minimum configuration system in figure 3 or the reduced minimum configuration shown in figure 14. The entire optical system including the fibre and the fibre components is manufactured in house and the modulation is derived from a simple PZT disc operating at the loop transit frequency (for most designs) around 135 kHz. The detection electronics uses relative harmonic amplitudes as the means for phase detection and system stabilization. The necessary stable, relatively broadband (1 MHz or thereabouts) analogue systems required careful design. The performance of these gyroscopes includes dynamic range capability up to a few thousand degrees a second, with random walk levels varying between around 0.1 and 20 degrees per hour per root hertz. For many of the less demanding applications (with higher values of random walk) cost is paramount and the reduced configuration approach shown in figure 14 has been successfully applied. In common with most fibre gyroscopes, these systems are all tested to a full MIL specification with the necessary thermal and vibrational tolerance. Consequently coil design, optical system mounting procedures and packaging are extremely important—and time consuming—factors in realizing the complete system [21].

Ixsea in contrast have concentrated on the high performance and therefore much higher unit cost sector. Ixsea's gyroscopes are closed loop with an integrated optic head end (figure 15). Furthermore, the signal detection and processing is now predominantly digital. The majority of the essential components are procured externally with Ixsea addressing the complex issues of system design including not only the optics but also the electronics and the mechanical packaging. The 'PHINS' inertial navigation system complies with the industry standard one mile per hour requirements [2] and the performance has now been driven to the 10^{-3} deg per hour category for space and Earth observation systems. Another major application

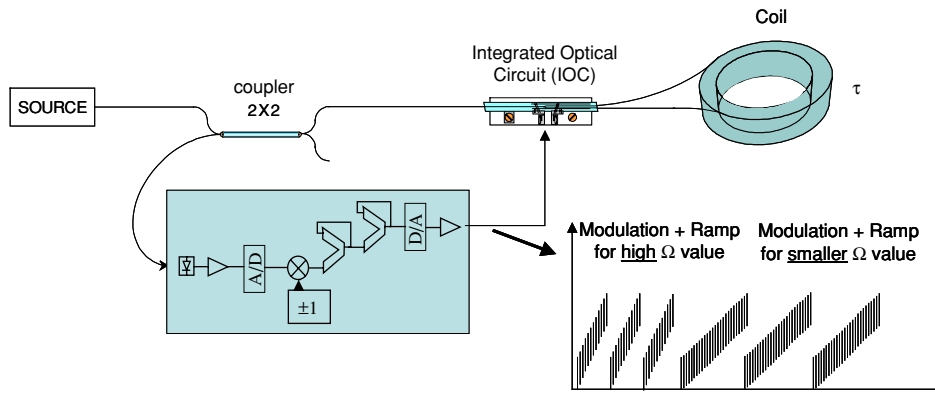


Figure 15. The Ixsea gyroscope architecture (reproduced from [22] with the authors' permission).

sector is subsea navigation including a recently proposed product known as GAPS (global acoustic positioning system) which combines underwater inertial navigation with acoustic positioning [22].

These two products technologically contrast significantly. They do, however, share a common design philosophy in that one basic technological platform is used to provide the variety of both product lines and is also the probable basis of future developments. They have another feature in common. Both have been driven very significantly by a particular individual. In KVH, the long term vision and persistence came from Dick Dyott, whilst in Ixsea the foundations were laid predominantly through the inspiration of Hervé Lefèvre—the author of the text in [6]. Interestingly both KVH and Ixsea are also relatively small specialized companies.

The traditional major navigational system suppliers dominate the remainder of the fibre gyroscope activity. Again the two basic architectures based on a minimum configuration optical system dominate the activities and the applications address a mix of stabilization and navigational requirements. The fibre gyroscope has now most definitely found its niche. It is a precision instrument which in manufacturing terms is relatively simple to assemble and is based upon a relatively modest specification component set. With its rather more complex active Sagnac equivalent, the laser gyroscope, it is beginning to predominate in particular sectors of the rotational measurement system market—though mechanical gyroscopes are still sold in large volumes. It is difficult to estimate the actual volume of fibre gyroscope sales, though an educated guess puts this in the few hundred thousand units per annum region worldwide. The details are however jealously guarded. It is though without doubt the most successful, to date, of fibre optic sensor technologies in terms of both technical achievements and market penetration. It is an area where a fibre optic device is simply taken for granted as a perfectly normal product. The development process took time from the first demonstrations in the mid 1970s to the first product in the mid 1990s but precision instruments is an area in which patience is essential. Its success is self-evident.

4. Other things to do with a Sagnac interferometer

There are many, most of which use the inherent reciprocity characteristics to generate an interferometer which normally

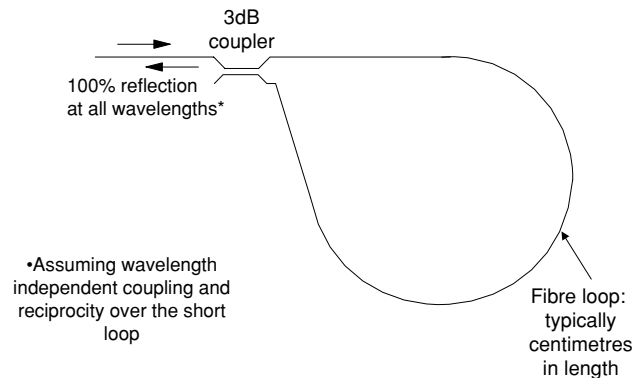


Figure 16. The optical fibre loop mirror.

works on or around the zero fringe point. We shall briefly mention the three most important, namely the loop mirror, the Sagnac current monitor and Sagnac configurations for use as hydrophones and geophones.

4.1. The fibre loop mirror

The principle of the loop mirror is very simple and is shown in figure 16. Essentially the loop mirror comprises a 50:50 coupler with the output tails spliced together or alternatively a loop of fibre with a coupler fused on one end and adjusted for zero output on the free port. The loop in the loop mirror is typically very short—centimetres in length—and therefore its sensitivity as a rotation-measuring device is minimal. Consequently, provided other effects do not come into play, all the incident light is reflected back down the input fibre regardless of wavelength and operating conditions. The loop mirror is consequently a very high reflectivity broadband reflector which since it was first mooted in 1988 [23] has found multiple uses, such as for example the terminating mirror in a fibre laser or dual pass amplifier. It has also found applications as mirrors in routing systems for all optical networks, delay and recombine processing networks and a plethora of other systems.

There are of course multiple variations on the basic loop mirror theme of which the most common uses polarization phenomena and birefringence within the fibre loop. The basic idea is simple. Suppose for example we insert linearly polarized light into the loop and the birefringence in the loop

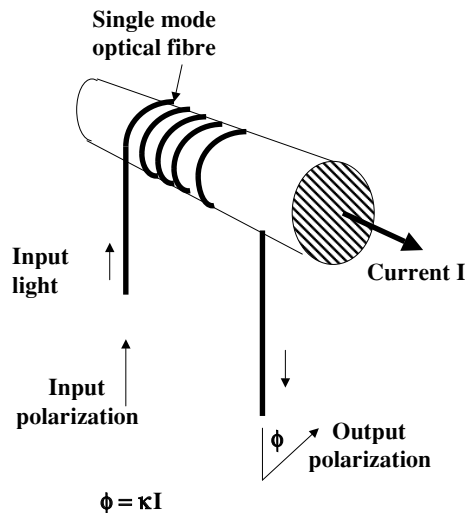


Figure 17. Faraday rotation current monitor: κ is the Verdet constant.

is such that the emerging beams are orthogonally polarized. There will then be no interference and equal amounts of light will emerge from each port. We could also introduce birefringence such that the two states on emerging are 180° out of phase in which case all the light emerges from the free port and none from the input port. This basic concept has been extensively analysed [24] and by, in effect, controlling the beat length within the fibre loop, it is feasible to make a range of wavelength selective loop mirror filters. This filter is nothing more than a controlled deviation from the basic conditions necessary to meet Lorenz reciprocity.

Nonlinear phenomena can also be exploited to introduce similar controlled deviations from reciprocity. In these devices optical bistability can be induced within the loop mirror at sufficiently high power levels to produce a very fast optical switch [25]. These fast optical switches use the cross phase modulation induced by the optical Kerr effect to induce a differential delay in the two counter propagating directions.

The loop mirror has in practice proved to be an extremely useful and somewhat unexpected by-product of the research into the fibre Sagnac interferometer which was originally stimulated by its potential use as a rotation sensor.

4.2. The optical fibre Sagnac current sensor

Faraday rotation has been extensively used as the basic principle for optical fibre current monitoring (figure 17). The principle is simple. A linear state of polarization is introduced into a non-birefringent fibre loop and this state of polarization rotates through an angle which is dependent upon the line integral of the magnetic field along the propagation path around the loop—in other words upon the current enclosed within the loop [26]. Faraday rotation is however a misnomer. A more fundamental way of viewing Faraday rotation is as a magnetic field induced circular birefringence. Consequently if circularly polarized light is introduced into a fibre current monitor of the type shown in figure 17 there will be a current-dependent phase modulation on the circularly birefringent light rather than a rotation in output state. Furthermore this phase delay is non-reciprocal so that the change in phase in

one direction is equal and opposite to the phase in the other. Clearly this could be detected using the Sagnac configuration.

This basic principle has found some application as a fibre Sagnac current monitor [27]. However for successful implementation circularly birefringent fibre is essential for most implementations. This ensures that the circular state is maintained throughout the fibre propagation length within the Sagnac loop and as in the normal Sagnac interferometer single polarization state propagation in both directions is essential. Optical fibre sensing technology is at present dominated by straightforward polarization rotation devices but there are a few examples of prototype systems using the Sagnac principle¹.

4.3. Sagnac interferometers as hydrophones and geophones

We have already seen that the Sagnac interferometer is sensitive to transient asymmetric disturbances. A dynamic perturbation occurring at one end of the fibre loop changes in the time interval between the arrival of counter propagating beams and therefore introduces a non-reciprocity. Indeed this is the basic principle of the detection modulation schemes.

It is then a relatively short step to change this phenomenon into a useful sensing system by masking one half of the loop from the physical field of interest and exposing the other. The more rapidly the dynamic field changes the more sensitive the interferometer becomes to a particular phase deviation until reaching a maximum at a frequency closely related to the transit time frequency which we have discussed already. Bringing the response into the acoustic frequency band requires long fibre loops and figure 18 shows the frequency response of a Sagnac hydrophone for partially marked loop lengths of 20 and 40 km—far longer than would normally be used in a gyroscope configuration [28].

The immense benefit of the Sagnac-based hydrophone and geophone systems is their operation on, by definition, the zero fringe of the interferometer. This has a substantial impact upon the noise performance. Conventionally Mach-Zehnder hydrophones attempt to balance the differential path lengths in the two arms such that phase to amplitude noise conversion from the optical source is minimized. In the Sagnac interferometer this phase to noise amplitude noise conversion is for all practical purposes zero. The original concept of the Sagnac-based hydrophone was first explored some time ago but more recently Sagnac interferometer arrays for geophone and hydrophone systems have been extensively investigated and realized as engineering prototypes [29]. The basic configuration (an example is shown in figure 19) for the Sagnac array of sensors is in practice quite complex and requires careful design. In particular, acoustic pick up from areas other than the sensor coils and polarization fading are important issues. The depolarized system shown in figure 19 copes with the latter and careful design of delay coils and loop timing can significantly minimize the former. Whilst considerable expertise has been accumulated on Mach-Zehnder interferometer arrays, the benefits of the Sagnac-based device are extremely persuasive and include far greater

¹ Commercial fibre optic current monitors are available from, among others, ABB, Nxt Phase, Siemens, Toshiba Power Systems.

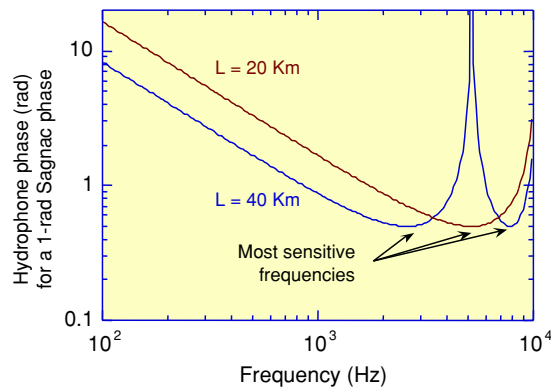


Figure 18. Frequency response for the Sagnac loop when operated as a fibre optic acoustic detector. The Sagnac loop is most sensitive when the acoustic frequency f satisfies: $2\pi f \tau_t = \pi/2$ (after [29] with the authors' permission).

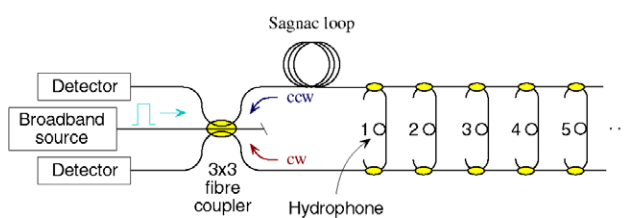


Figure 19. A fibre Sagnac hydrophone array. This is based on a 3×3 coupler providing a $\pm 2\pi/3$ phase bias and a single multi-sensor TDM loop (after [29] with the authors' permission).

dynamic range and immunity to signal fading and polarization drift. These issues are in the process of being resolved in an engineering context and extensive array field trials are imminent.

4.4. And many others

The dynamic sensitivity of the Sagnac loop has found numerous other potential applications. These include strain field sensing and distributed sensing architectures, not to mention security monitoring, intrusion detection and secure communications networks [30]. Many of these possible applications are due to Eric Udd, of Blue Road Research, Portland, Oregon, and have been described extensively in the patent literature. There is though a common theme—namely that the applications all rely upon some mechanism to perturb the reciprocity of the interferometer whilst minimizing the influence of potential interfering influences.

5. Concluding comments

The fibre optic Sagnac interferometer has proved to be arguably the most successful example to date of optical fibre sensor technology. The configuration has many benefits starting with operation on the zero fringe with its attendant inherent immunity to phase to amplitude noise conversion—a phenomenon which affects all non-zero path difference interferometers. Further, as Hervé Lefèvre pointed out some years ago—it continues to defy Murphy's law [22]. There are indeed many examples of phenomena where correcting for one

influence, for example backscatter, by using the transit time modulation frequency, corrects for others too such as the onset of spurious amplitude modulation on the light propagating within the fibre loop.

When used as a gyroscope the fibre Sagnac interferometer has proved to have enormously versatile properties. It can be configured in a relatively low cost version with modest performance (though sometimes not modest enough) to begin to compete effectively with MEMS technologies and similar systems. At the high performance end millidegrees per hour random walk noise are now feasible and at this level space missions and long-term submarine voyages become a possibility. The fibre gyroscopes at these extremes are very different, going from short open loop coils to long closed loop configurations through a host of intermediate stages.

It is though the broad utility of the Sagnac interferometer which continues to fascinate the technology community. The loop mirror has become the ubiquitous reflector. Furthermore, at high power levels it can become an optical switch. Additionally, if it is twisted or made birefringent the mirror turns into a filter. Using other non-reciprocal effects in sensing has also progressed from curiosity to real device. Sagnac detection of Faraday rotation (more correctly non-reciprocal current induced circular birefringence...) gives a new generation of current sensors which complement the loop wound direct Faraday rotation systems. Time-dependent phenomena have also been very effectively detected using the Sagnac interferometer and recent progress with hydrophones and geophones especially in networked arrays has taken this from a laboratory concept a decade or so ago into field prototype evaluation systems.

What next? Some have suggested active Sagnac systems with Brillouin or Raman amplifiers in the loop to further enhance detection sensitivity. However, the obvious comment here is that gain in the loop implies non-reciprocity and very small variations in gain could give large non-reciprocal phases. Consequently, the high sensitivity could well be offset by significant non-reciprocal drift. Furthermore, the conventional interferometer gyroscope has already demonstrated its worth in extremely demanding applications. Other prospects have been proposed. A fibre Sagnac gravitational telescope based upon hundreds of kilometres of fibre configured in a slow wave structure to detect circularly polarized gravitational waves is one such idea [31]. For sure the sensitivities which have been achieved and the prospects for ingenious gain systems within the loop could open this up once again as an intriguing long term scientific and technological challenge. Quite who might fund it is a different debate.

The fibre Sagnac interferometer then has all the ingredients. It combines scientific elegance with technological practicality. It has provided intellectual stimulus and fascination for two, perhaps three, generations of fibre optic research and development engineers and it will continue for sure to contribute a vital role in guidance systems and optical fibre networks for a considerable time to come.

Acknowledgments and dedication

Over the past 20 or so years, I have been fortunate to meet and work with many of the founding influences in fibre Sagnac

interferometer research and development, and learnt much from these encounters. There are many, including Hervé Lefèvre, Ralph Bergh, Chap Cutler and John Shaw, Michel Digonnet, Eric Udd, Shaul Ezekiel, Robert Smith, John Dakin, Kazuo Hotate, Kjell Bløtekjær and doubtless many, many others. There have been the students and post docs—Ian Giles with whom I wrote a fibre gyro review for this journal many years ago, James Mackintosh, Duncan Platt, Jin Wei and others. The anonymous reviewer too gave useful and constructive comments on the draft, resulting in, hopefully, a clearer and more accurate review.

Most of all I learnt much from a close involvement with Dick Dyott and the Andrew/KVH gyro development programme. Dick helped in the preparation of this review and sadly died just before the review was completed. Dick was a wonderfully inventive engineer, an intuitive technologist and an inspiration and mentor to all who knew him. He was also a great friend with a mischievous humour and deep humanity. I would like to dedicate this review to Dick's memory as a small personal tribute to his contributions to science and technology over a career approaching 60 years and to his continued humanity.

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